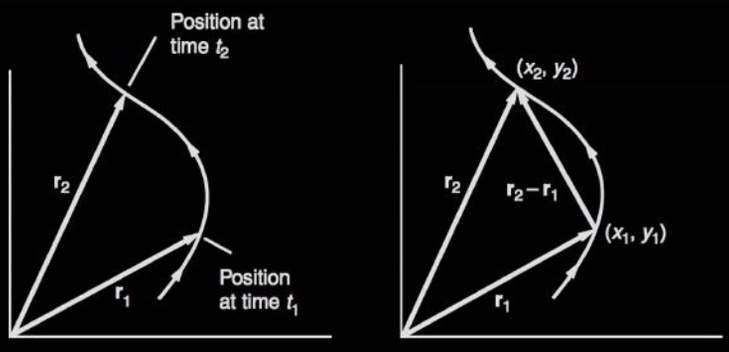
Velocity And Acceleration In 2D & 3D Motion:

Recall that during our study of vectors we had defined position vector of a particle as a vector joining origin to the location of the particle and displacement as the change in position vector.



We can now extend the definition of velocity and acceleration in general motion in space as:

$$\vec{v}_{avg} = \frac{\Delta \vec{r}}{\Delta t} = \frac{\Delta \vec{s}}{\Delta t} = \left[(x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j} + (z_2 - z_1)\hat{k}\right] / \Delta t.$$

Here the particle has moved from (x_1, y_1, z_1) to (x_2, y_2, z_2).

Now instantaneous velocity is nothing but average velocity for infinitesimal time thus:

$$\vec{v} = \frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j} + \frac{dz}{dt}\hat{k} = \frac{d\vec{r}}{dt}$$

Average acceleration is defined as average rate of change of velocity. Thus:

$$\vec{a}_{avg} = \frac{\Delta \vec{v}}{\Delta t} = \left(\frac{\vec{v} - \vec{u}}{\Delta t}\right),$$ here $\vec{v}$ is the velocity at the end of the time interval and $\vec{u}$ is the initial velocity. Thus vectorially we can also write:

$$\vec{a}_{avg} = \left[(v_x - u_x)\hat{i} + (v_y - u_y)\hat{j} + (v_z - u_z)\hat{k}\right] / \Delta t.$$

Now instantaneous acceleration is nothing but average acceleration for an infinitesimal time interval. Thus:

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{dv_x}{dt}\hat{i} + \frac{dv_y}{dt}\hat{j} + \frac{dv_z}{dt}\hat{k}$$

$$= \frac{d^2x}{dt^2}\hat{i} + \frac{d^2y}{dt^2}\hat{j} + \frac{d^2z}{dt^2}\hat{k} = \frac{d^2\vec{r}}{dt^2}$$

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EXAMPLE: The position of a particle is given as $\vec{r} = A[e^{\alpha t}\hat{i} + e^{-\alpha t}\hat{j}]$, find the velocity and acceleration of particle at general time and sketch its trajectory.

Solution: $x = Ae^{\alpha t}$, $y = Ae^{-\alpha t} \Rightarrow v_x = dx/dt = A\alpha e^{\alpha t}$

$v_y = dy/dt = -A\alpha e^{-\alpha t}$, $a_x = \frac{dv_x}{dt} = A\alpha^2 e^{\alpha t}$,

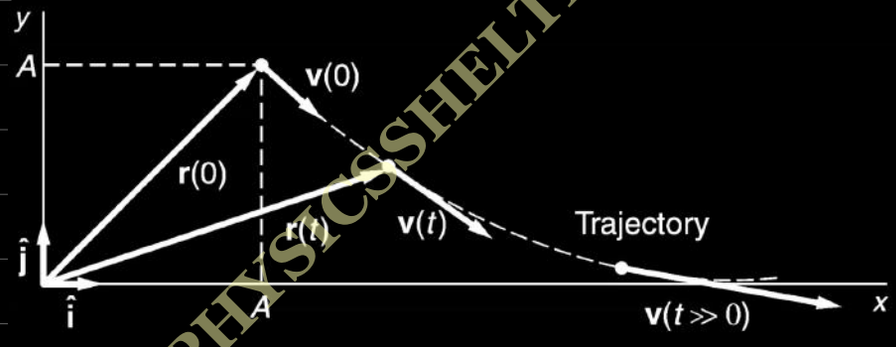
$a_y = dv_y/dt = A\alpha^2 e^{-\alpha t}$ thus:

$\vec{v} = A\alpha[e^{\alpha t}\hat{i} - e^{-\alpha t}\hat{j}]$ and $\vec{a} = A\alpha^2[e^{\alpha t}\hat{i} + e^{-\alpha t}\hat{j}]$ Ans.

Now $xy = A^2 \Rightarrow y = A^2/x$.

Also $x = Ae^{\alpha t}$, $y = Ae^{-\alpha t}$

thus @ $t=0$, particle starts from (A, A) and at $t=\infty$ it goes to $(\infty, 0)$.



→ Velocity is always along tangent to the path.

DIFFERENTIAL EQUATIONS GOVERNING 2-D AND 3D MOTION

Just as we had differential equations governing 1D motion, we have similar equations governing 2D and 3D motions along each axis thus:

$$v_x = \frac{dx}{dt} = \frac{ds_x}{dt}, \quad v_y = \frac{dy}{dt} = \frac{ds_y}{dt}, \quad v_z = \frac{dz}{dt} = \frac{ds_z}{dt}$$

Similarly:

$$a_x = \frac{dv_x}{dt} = v_x\frac{dv_x}{dx}, \quad a_y = \frac{dv_y}{dt} = v_y\frac{dv_y}{dy}, \quad a_z = \frac{dv_z}{dt} = v_z\frac{dv_z}{dz}$$

These equations can be appropriately integrated or differentiated

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to get the required relations in various kinematic variables.

EXAMPLE: IRODOV: 1.26:

A point moves in the plane xy according to the law $x = a\sin\omega t$, $y = a(1-\cos\omega t)$, where a and ω are positive constants, Find :
(a) the distance s traversed by the point during the time τ;
(b) the angle between the point's velocity and acceleration vectors.

Solution: $v_x = dx/dt = a\omega\cos\omega t$, $v_y = dy/dt = a\omega\sin\omega t$.

Now speed $= dl/dt = \sqrt{v_x^2 + v_y^2} = a\omega$, thus speed of the particle is constant $\therefore$ distance travelled in time τ, $l = a\omega\tau$ Ans.

Now accelerations:

$w_x = dv_x/dt = -a\omega^2\sin\omega t$, $w_y = dv_y/dt = a\omega^2\cos\omega t$

thus: $\vec{w} = a\omega^2(-\sin\omega t\,\hat{i} + \cos\omega t\,\hat{j})$, $\vec{v} = a\omega(\cos\omega t\,\hat{i} + \sin\omega t\,\hat{j})$.

One can readily see that $\vec{v}\cdot\vec{w} = 0$, thus angle between velocity and acceleration must be 90° (recall that dot product of mutually perpendicular vectors is zero).

EXAMPLE: IRODOV: 1.27:

A particle moves in the plane xy with constant acceleration $\vec{w}$ directed along the negative y axis. The equation of motion of the particle has the form $y = ax - bx^2$, where a and b are positive constants. Find the velocity of the particle at the origin of coordinates.

Solution: $y = ax - bx^2$

$$\Rightarrow \frac{dy}{dt} = a\frac{dx}{dt} - 2bx\frac{dx}{dt} \quad (1)$$

$$\Rightarrow \frac{d^2y}{dt^2} = a\frac{d^2x}{dt^2} - 2b\left[x\frac{d^2x}{dt^2} + \left(\frac{dx}{dt}\right)^2\right] = -w \quad (2)$$

but $d^2x/dt^2 = 0$ ($\because$ acceleration is only along $-y$) thus:

$$-w = -2b(dx/dt)^2 \Rightarrow dx/dt = \sqrt{w/2b} = v_x. \quad (3)$$

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Now using ③ in ①: At the origin:

$$\frac{dy}{dt} = a \times v_x - 2b \times 0 \times v_x = a v_x = a\sqrt{w/2b}$$

thus $\vec{v} = \sqrt{w/2b}\,\hat{i} + a\sqrt{w/2b}\,\hat{j}$, speed, $v = \sqrt{(1+a^2)\,w/2b}$ Ans.

EXAMPLE: IRODOV: 1.34:

A balloon starts rising from the surface of the Earth. The ascension rate is constant and equal to v_0. Due to the wind the balloon gathers the horizontal velocity component $v_x = ay$, where a is a constant and y is the height of ascent. Find how the following quantities depend on the height of ascent;

(a) the horizontal, drift of the balloon $x(y)$;

(b) the total, tangential, and normal accelerations of the balloon.

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Solution: Assuming initial position as origin, $y = v_0 t$, $v_x = ay$

$\Rightarrow v_x = a v_0 t = dx/dt \Rightarrow x = \int a v_0 t\,dt = \frac{a}{2} v_0 t^2$

Eliminating t from $x(t)$, $y(t)$; $x = \frac{a v_0}{2}[y/v_0]^2 = \frac{ay^2}{2v_0}$ Ans.

Now $y = \sqrt{2v_0 x/a}$. $w_y = dv_y/dt = 0$, $w_x = dv_x/dt = a v_0$

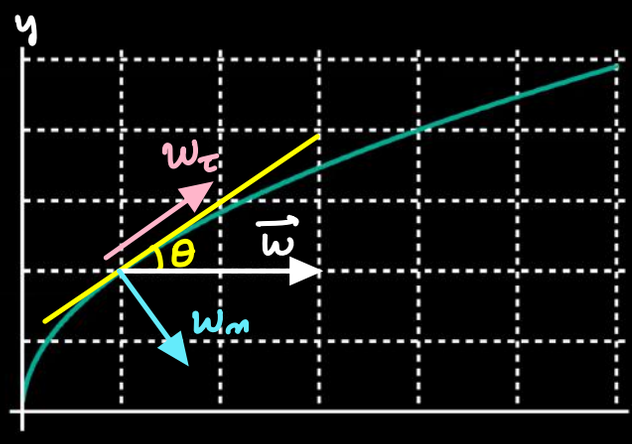
thus $\vec{w} = a v_0 \hat{i}$ this is total acceleration. Now to find its tangential and normal component, we need to find tangential unit vector.

Now unit vector along tangent,

$\hat{\tau} = \cos\theta\,\hat{i} + \sin\theta\,\hat{j}$, Now,

$\tan\theta = dy/dx \Rightarrow \cot\theta = dx/dy$

$\Rightarrow \cot\theta = ay/v_0$



$\Rightarrow \sin\theta = v_0/\sqrt{v_0^2 + (ay)^2}$

$\cos\theta = ay/\sqrt{v_0^2 + (ay)^2}$

Now tangential acceleration, $\vec{w}_\tau = (\vec{w}\cdot\hat{\tau})\hat{\tau}$ thus,

Tangential acceleration $= \left[a^2 v_0 y / \sqrt{v_0^2 + a^2 y^2} \right] \hat{\tau} = \vec{w_t}$

Normal acceleration, $w_n = w \sin\theta = a v_0^2 / \sqrt{v_0^2 + a^2 y^2}$ Ans.

Think: What is the radius of curvature of the path at a general altitude y?

Solution: $r = \dfrac{\left[1 + (dy/dx)^2\right]^{3/2}}{d^2y/dx^2}$

Now $x = ay^2/2v_0$

$\Rightarrow dx/dy = ay/v_0$

$\Rightarrow dy/dx = v_0/ay$.

Now $\dfrac{d^2y}{dx^2} = \dfrac{v_0}{a} \cdot \dfrac{-1}{y^2} \cdot (dy/dx)$

$= \dfrac{-v_0}{ay^2} \times \dfrac{v_0}{ay} = \dfrac{-v_0^2}{a^2y^3} \Rightarrow r = -\dfrac{\left[1 + (v_0/ay)^2\right]^{3/2} a^2 y^3}{v_0^2}$ Ans.

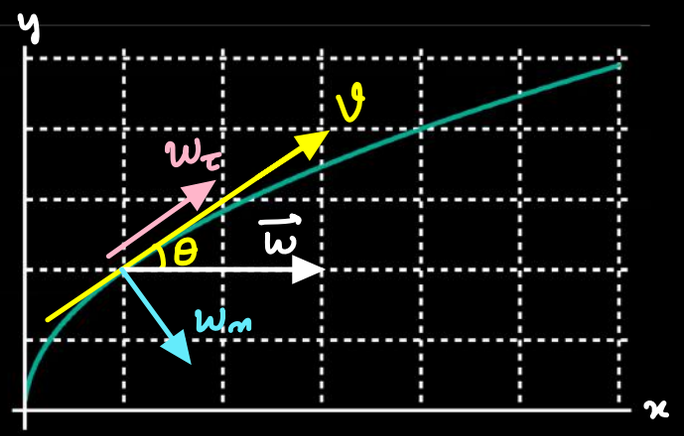
→ Later on we will derive that $r = v^2/w_n$

Relation Between Tangential Acceleration And Rate Of Change Of Speed:

Let a particle move in the x-y plane with a velocity $\vec{v} = v_x \hat{i} + v_y \hat{j}$. The velocity always along tangent of the path. Thus unit vector along tangent:

$$\hat{\tau} = (v_x \hat{i} + v_y \hat{j}) / \sqrt{v_x^2 + v_y^2}$$

Now $\vec{a} = \dfrac{d\vec{v}}{dt} = a_x \hat{i} + a_y \hat{j}$, the



component of acceleration along the tangential direction (or velocity vector) is called tangential acceleration. Thus:

$$a_t = \vec{a} \cdot \hat{\tau} = (a_x v_x + a_y v_y) / \sqrt{v_x^2 + v_y^2} \quad (1)$$

Now speed of the particle, $v = \sqrt{v_x^2 + v_y^2}$

thus rate of change of speed:

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$$\frac{dv}{dt} = \frac{1}{2\sqrt{v_x^2 + v_y^2}} \cdot \left(2v_x \, dv_x/dt + 2v_y \, dv_y/dt\right)$$

$$\Rightarrow dv/dt = \left(v_x a_x + v_y a_y\right) / \sqrt{v_x^2 + v_y^2} \quad (2)$$

From (1) & (2) we can conclude that tangential acceleration is nothing but rate of change of speed of the particle. Thus:

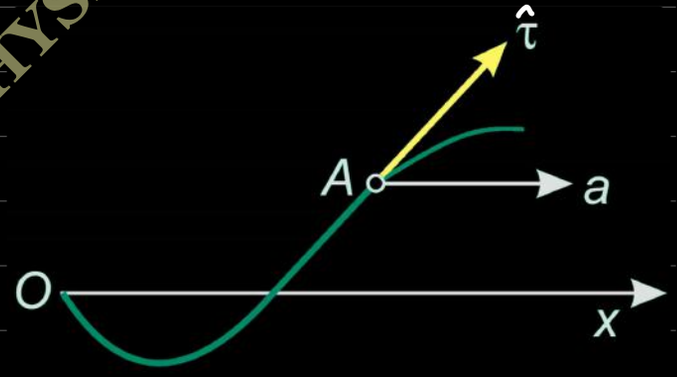
$$\boxed{a_t = dv/dt = v \, dv/dl}$$

→ A positive tangential acceleration happens when angle between $\vec{a}$ and $\vec{v}$ is acute and it means speed is increasing

→ A negative tangential acceleration means that speed is decreasing (happens when angle between $\vec{a}$ and $\vec{v}$ is obtuse).

EXAMPLE : IRODOV : 1.36 :

A particle moves in one direction along a given trajectory with a tangential acceleration $w_\tau = \vec{a} \cdot \hat{\tau}$ where $\vec{a}$ is a constant vector parallel to x-axis, and $\hat{\tau}$ is tangential unit vector (along velocity). Find the speed as a function of x coordinate assuming negligible speed at $x = 0$.



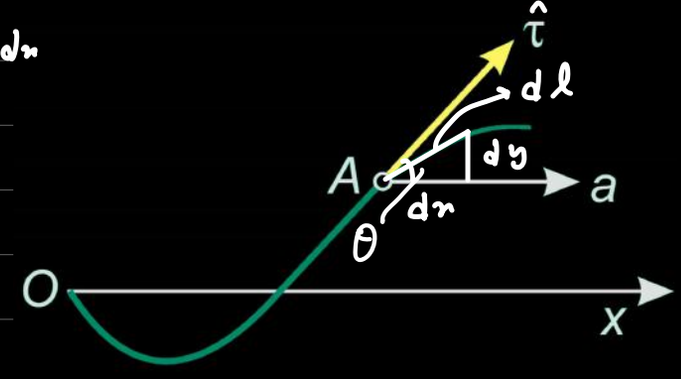
Solution: $a\cos\theta = dv/dt = v \, dv/dl$

$\Rightarrow v \, dv = a \, dl \cos\theta$, Now $dl\cos\theta = dx$

$\Rightarrow v \, dv = a \, dx$

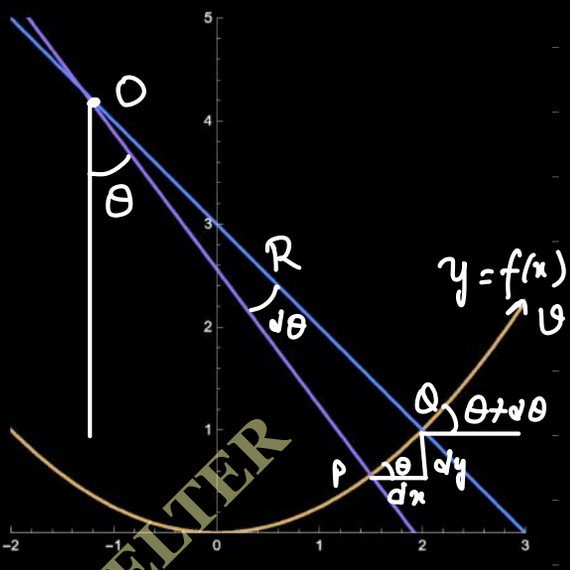
$$\Rightarrow \int_0^v v \, dv = \int_0^x a \, dx$$

$\Rightarrow v^2 = 2ax \Rightarrow v = \sqrt{2ax}$ Ans.

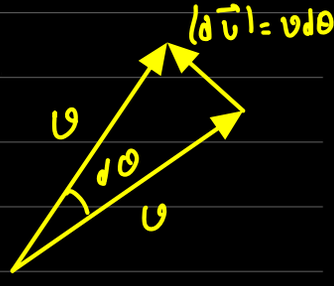


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Relation Between Normal Acceleration And Radius Of Curvature:



Recall that tangential acceleration is rate of change of speed of a particle. This means that if a particle is moving with constant speed along a curve, its acceleration (if any) must be normal to the curve. Now let a particle move with constant speed along a curve. We consider infinitesimal time in which a particle moves from P to Q (see adjacent figure). Since during this time interval, radius vector has turned through an angle $d\theta$, the tangent to the curve must also have turned through an angle $d\theta$. Thus velocity vector must also have turned through an angle $d\theta$. Thus change in velocity vector is obtained as shown in the adjacent diagram. For very small $d\theta$, arc length and chord length for an arc of radius v must be same. Thus change in velocity vector is $v\,d\theta$ in a direction $\perp$ to $\vec{v}$ ie along the normal to the curve (∵ $\vec{v}$ is along tangent to the curve). Thus acceleration $\vec{a} = \frac{d\vec{v}}{dt} = \frac{v\,d\theta\,\hat{n}}{dt}$ (1)



where $\hat{n}$ represents normal to the curve in the direction of centre of curvature. Now $dl = v\,dt = R\,d\theta \Rightarrow d\theta/dt = v/R$ (2)

Now using (2) in (1),

$$\vec{a} = \frac{v^2}{R}\hat{n}$$, thus acceleration of a particle moving along a curve with uniform speed is $\frac{v^2}{R}$ along normal to the curve. In case the speed is not constant, $\frac{v^2}{R}\hat{n}$ represents the normal component of acceleration i.e.:

$a_n = v^2/R$, thus radius of curvature can be written as: $\boxed{R = v^2/a_n}$ (4)

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Note that radius of curvature is purely a shape dependent geometrical parameter of the curve being traversed so it does not depend on how fast or slow the curve is being travelled (or whether traveling speed is constant or changing). Thus for finding radius of curvature, we can always assume a constant speed and then use formula ④.

EXAMPLE: IRODOV: 1.41:

A point moves in the plane so that its tangential acceleration $w_\tau = a$, and its normal acceleration $w_n = bt^4$, where a and b are positive constants, and t is time. At the moment $t = 0$ the point was at rest. Find how the curvature radius R of the point's trajectory and the total acceleration w depend on the distance covered ℓ.

Solution: $w_\tau = v\, dv/d\ell = a \Rightarrow v\,dv = a\,d\ell$

$$\Rightarrow \int_0^v v\,dv = \int_0^\ell a\,d\ell \Rightarrow v^2 = 2a\ell \quad ①$$

also $w_\tau = dv/dt = a \Rightarrow \int_0^v dv = \int_0^t a\,dt \Rightarrow v = at$ ②

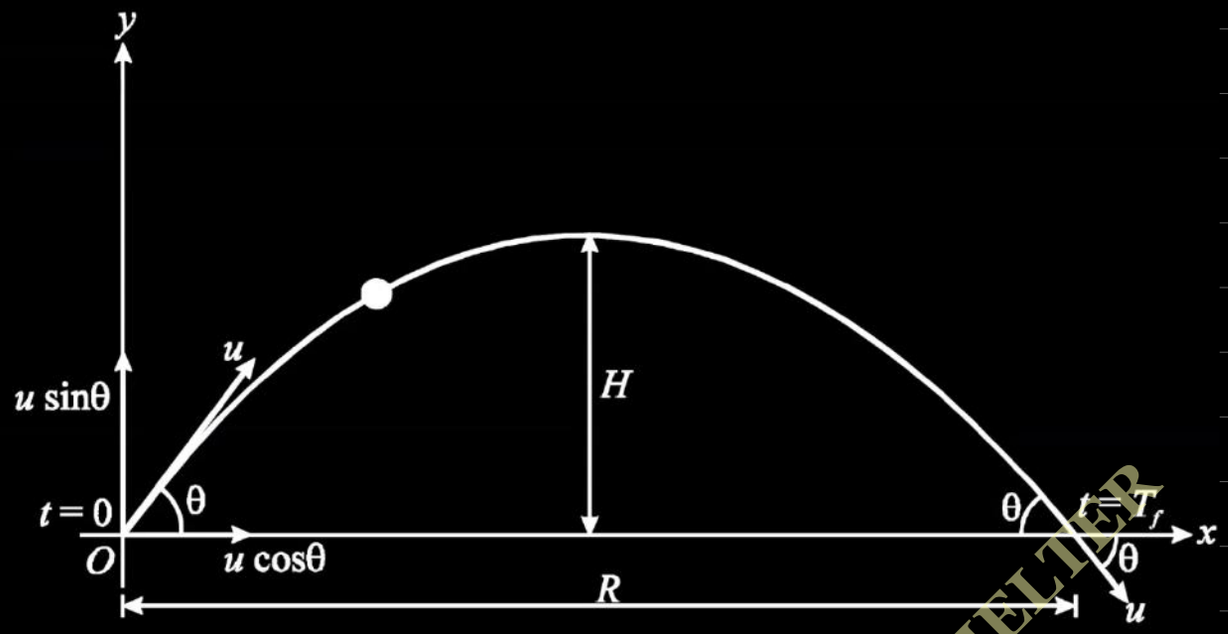
Now $w_n = bt^4 = b(v/a)^4 = \frac{b}{a^4} \times 4a^2\ell^2 = 4b\ell^2/a^2$ ③

Now $R = v^2/a_n = 2a\ell / (4b\ell^2/a^2) = a^3/2b\ell$ Ans.

Now total acceleration: $w = \sqrt{w_n^2 + w_\tau^2} = \sqrt{a^2 + (4b\ell^2/a^2)^2}$ Ans.

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Projectile Motion:



Consider a stone thrown from origin with a speed u at an angle θ. Until its fall its acceleration is due to gravity alone (ignoring air drag), thus it has a uniformly accelerated motion in vertical direction, and in the horizontal direction, its speed remains unchanged. We are interested in deriving several results related to its motion.

Here, $u_x = u\cos\theta$, $u_y = u\sin\theta$, $a_x = 0$, $a_y = -g$.

Time Of Flight For Ground To Ground Projectile:

$S_y = u_y t + \frac{1}{2} a_y t^2$ now @ $t = T$, $S_y = 0 \Rightarrow u\sin\theta T - \frac{1}{2} g T^2 = 0$

$\Rightarrow$ $\boxed{T = 2u\sin\theta/g}$

Range Of A Ground To Ground Projectile:

$S_x = u_x t + \frac{1}{2} a_x t^2$, now at $t = T$, $S_x = R$ thus:

$R = u\cos\theta \times 2u\sin\theta/g + 0 = u^2 \sin 2\theta/g$. thus:

$\boxed{R = u^2 \sin 2\theta / g}$

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→ For max range, $\sin 2\theta = 1 \Rightarrow \theta = 45^\circ$ and $R_{max} = u^2/g$.

→ Range at complimentary angles is same, this can be easily seen thus:

$$R_{90^\circ-\theta} = u^2 \sin 2(90^\circ-\theta)/g = u^2 \sin[180^\circ - 2\theta)/g$$
$$= u^2 \sin 2\theta/g \quad \text{thus } R_\theta = R_{90^\circ-\theta} \quad (\because \sin 180^\circ - 2\theta = \sin 2\theta).$$

Maximum Height Above Point Of Projection:

$v_y^2 - u_y^2 = 2a_y s_y$, now at $s_y = H$, $v_y = 0$ thus:

$$0^2 - u^2\sin^2\theta = 2(-g)H \Rightarrow \boxed{H = u^2\sin^2\theta/2g}$$

Trajectory Equation:

$x = u\cos\theta\, t$, $y = u\sin\theta\, t - \frac{1}{2}gt^2$ eliminating t:

$$y = u\sin\theta \times \frac{x}{u\cos\theta} - \frac{1}{2}g\cdot\left(\frac{x}{u\cos\theta}\right)^2 \Rightarrow y = x\tan\theta - \frac{1}{2}g\frac{x^2}{u^2\cos^2\theta} \text{ thus:}$$

$$\boxed{y = x\tan\theta - \frac{1}{2}g\frac{x^2}{u^2\cos^2\theta}}$$ ← Also true below ground level.

Range Form:

$$y = x\tan\theta\left[1 - \frac{x}{[2u^2\cos^2\theta\cdot\tan\theta/g]}\right]$$

$$\Rightarrow y = x\tan\theta\left[1 - \frac{x}{(u^2\sin 2\theta/g)}\right] \Rightarrow$$

$$\boxed{y = x\tan\theta\,[1 - x/R]}$$ ← Also true below ground level if projectile continues, R is range at ground level.

Speed At A General Flying Angle ϕ:

Since there is no accelerations in horizontal direction, the horizontal component of velocity always remains constant. Suppose the projectile is flying with a speed v at an angle ϕ then:

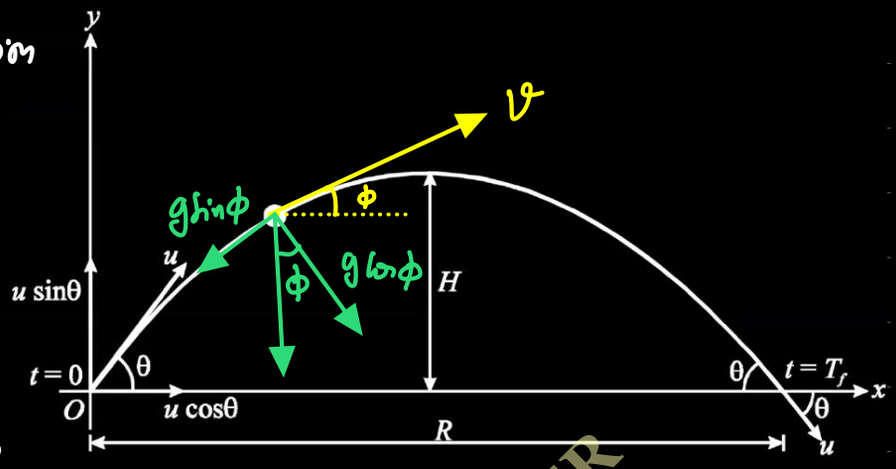
$$u\cos\theta = v\cos\phi \Rightarrow$$

$$\boxed{v = u\cos\theta/\cos\phi}$$

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Radius Of Curvature At A General ϕ :

Recall that in a previous section we had proved that radius of curvature is related to normal acceleration as: $R = v^2/a_n$, now consider a projectile moving at an angle ϕ with horizontal then, $a_n = g\cos\phi$ (see figure) and $v = u\cos\theta/\cos\phi$ (∵ derived in previous section), thus:



$$R = v^2/a_n = \frac{u^2\cos^2\theta}{\cos^2\phi}\cdot\frac{1}{g\cos\phi} = \frac{u^2\cos^2\theta}{g\cos^3\phi}$$ thus:

$$\boxed{R = \frac{u^2\cos^2\theta}{g\cos^3\phi}}$$

→ Minimum radius of curvature occurs at $\cos\phi = 1$ i.e $\phi = 0$ on the highest point of the trajectory.

→ Maximum radius of curvature occurs for $\phi = 0$ for a ground to ground projectile (∵ $\phi \le 0$). Thus maximum radius of curvature occurs at landing and launching points for a ground to ground projectile.

Example: Suppose a projectile has a maximum height H and a Range R, show that horizontal distance between points where it is at a height ηH is $R\sqrt{1-\eta}$.

Solution: $y = x\tan\theta\,(1 - x/R) \Rightarrow \frac{x^2\tan\theta}{R} - x\tan\theta + y = 0$

We need to find difference of its roots for a given y,

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Sum of roots $= -b/a = R$, product of roots $= c/a = \frac{yR}{\tan\theta}$

Now difference of roots, $x_2 - x_1 = \sqrt{(x_2+x_1)^2 - 4x_2x_1}$

$\Rightarrow x_2 - x_1 = \sqrt{R^2 - 4yR/\tan\theta} = R\sqrt{1 - \frac{4y}{R\tan\theta}}$

Now $y = \eta H = \eta u^2\sin^2\theta/2g$

$R = u^2\sin 2\theta/g \Rightarrow y/R = \eta\tan\theta/4$, thus,

$x_2 - x_1 = R\sqrt{1-\eta} \Rightarrow \boxed{\Delta x = R\sqrt{1-\eta}}$

Time to cross points of height ηH, $\Delta t = \Delta x/u\cos\theta = \frac{u^2\sin 2\theta}{g}\sqrt{1-\eta}\cdot\frac{1}{u\cos\theta}$

$\Rightarrow \boxed{\Delta t = T\sqrt{1-\eta}}$

EXAMPLE: For a given launching speed, maximum heights at complimentary angle of projection are H_1 and H_2. Find (a) launching speed (b) Range (c) Maximum range for given launching speed.

Solution: Let launch angles be θ and $90°-\theta$, then, $H_1 = u^2\sin^2\theta/2g$ and $H_2 = u^2\cos^2\theta/2g$. Now $\cos^2\theta + \sin^2\theta = 1$ thus: $H_1 + H_2 = u^2/2g$ thus:

$\boxed{u = \sqrt{2g(H_1+H_2)}}$

Now $R = \frac{u^2\sin 2\theta}{g} = \frac{2u^2\sin\theta\cos\theta}{g} = \sqrt{16\frac{u^2\sin^2\theta}{2g}\times\frac{u^2\cos^2\theta}{2g}}$

$\Rightarrow \boxed{R = 4\sqrt{H_1H_2}}$, Max range $= \frac{u^2\sin(2\times 45°)}{g} = \frac{u^2}{g}$

$\Rightarrow$ Max range $= 2(H_1+H_2)$, $\rightarrow \tan\theta = \sqrt{\frac{H_1}{H_2}}$ or $\sqrt{\frac{H_2}{H_1}}$

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EXAMPLE: For a certain projection velocity, T_1 and T_2 are times of flight for complimentary angles for ground to ground projection. Find (a) Projection angles (b) Range

Solution: Let θ and $90°-\theta$ be the projection angles then:

$$T_1 = 2u\sin\theta/g \quad \text{and} \quad T_2 = 2u\cos\theta/g \text{, thus,}$$

$$\boxed{\tan\theta = T_1/T_2}$$

, Also $R = \dfrac{2u^2\sin\theta\cos\theta}{g} = \dfrac{g}{2}\dfrac{2u\sin\theta}{g}\times\dfrac{2u\cos\theta}{g}$

$\Rightarrow R = g/2\ T_1T_2$ thus $\boxed{R = \frac{1}{2}g\,T_1T_2}$

EXAMPLE: A ball projected with $u = 10\,m/s$ at 30°, just clears two poles of height 1m each. Find the separation between the poles.

Solution:

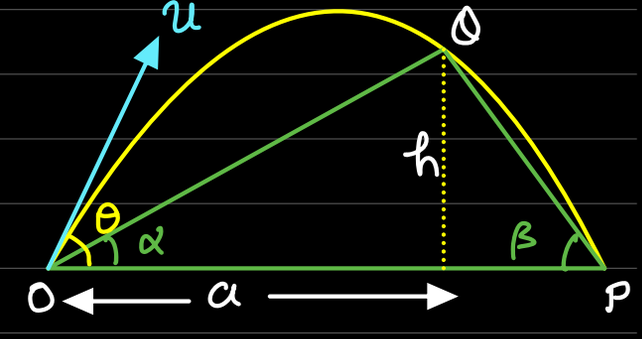
$$\Delta x = R\sqrt{1-\eta} = R\sqrt{1-y/H}$$

Now $R = u^2\sin 2\theta/g = 100\times\frac{\sqrt{3}}{2}\times\frac{1}{10} = 5\sqrt{3}\ m$, $H = \dfrac{u^2\sin^2\theta}{2g} = 100\times\frac{1}{4}\times\frac{1}{20}$

$\Rightarrow H = 5/4\ m$, thus $\Delta x = 5\sqrt{3}\sqrt{1-4/5} = \sqrt{15}\ m \approx 3.87\ m$

PROJECTILE ENVELOPING TRIANGLE WITH HORIZONTAL BASE:

Consider a projectile enveloping the shown triangle. Using trajectory equation:



$$h = a\tan\alpha = a\tan\theta\left(1-\frac{a}{R}\right) \quad (1)$$

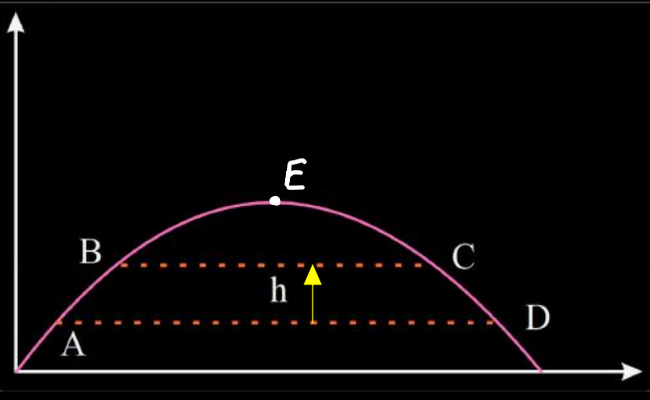
also, $h = (R-a)\tan\beta = a\tan\theta\left(1-\frac{a}{R}\right)$ (2)

$$\tan\alpha + \tan\beta = \left(1-\frac{a}{R}\right)\tan\theta\left[1+\frac{a}{R-a}\right] = \tan\theta\left[\frac{R-a}{R}\right]\left[\frac{R}{R-a}\right]$$

thus $\boxed{\tan\alpha + \tan\beta = \tan\theta}$

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EXAMPLE: It takes a projectile t_1 time to reach from A to D and t_2 time to reach from B to C. Find height h between the horizontal lines AD and BC



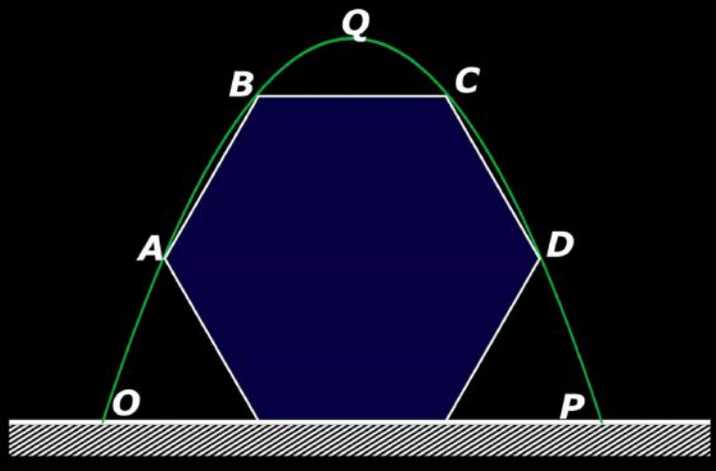
Solution Projectile motion is symmetric about the highest point. Thus time from highest point to C must be $t_2/2$ and that from highest point to D must be $t_1/2$. Now vertical distance between AD and BC:

$$h = \frac{1}{2} g (t_1/2)^2 - \frac{1}{2} g \left(\frac{t_2}{2}\right)^2 \quad (\because s_y = u_y + \frac{1}{2} a_y t^2).$$

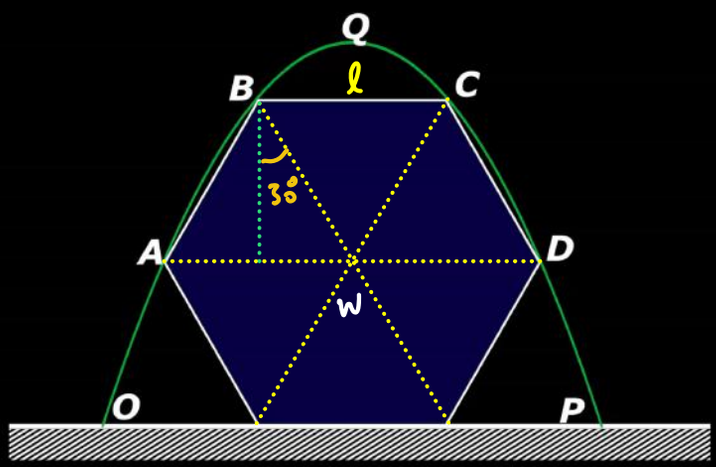
$$\Rightarrow \boxed{h = \frac{1}{8} g [t_1^2 - t_2^2]} \quad \text{Ans.}$$

EXAMPLE:

A particle is launched from ground to graze past four corners of a regular hexagon of side ℓ. Find (a) Range (b) maximum height (c) Angle of launch (d) launching speed.



Solution: Let $t_{BC} = \tau$, then $t_{AD} = 2\tau$ ∵ time is directly



proportional to horizontal distance (∵ horizontal speed is constant) then:

$$\frac{1}{8} g ((2\tau)^2 - \tau^2) = \ell\sqrt{3}/2 \quad (1)$$

Now let T be the time from

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O to P, then, $\frac{1}{8} g (T^2 - (2\tau)^2) = l\sqrt{3}/2$ ②

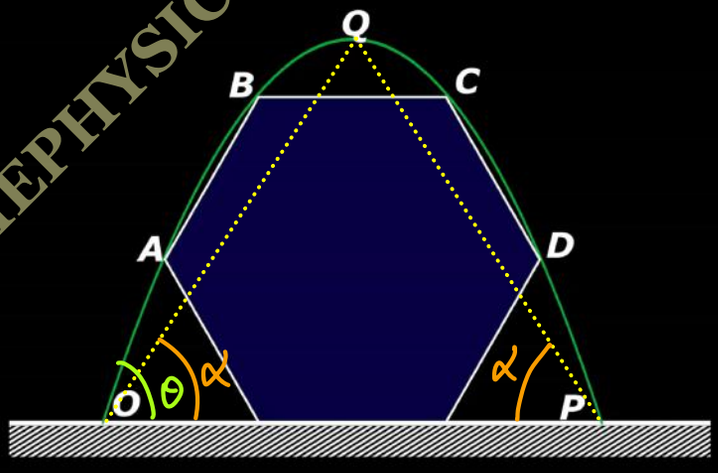
From ① & ②, $\frac{3\tau^2}{T^2 - 4\tau^2} = 1 \Rightarrow T = \sqrt{7}\,\tau$

Now since horizontal range is directly proportional to time of flight,
$R = \sqrt{7} \times BC = \sqrt{7}\, l$ Ans.

Now if max height is h, considering motion from Q to P:
$h = \frac{1}{2} g \left(\frac{\sqrt{7}\tau}{2}\right)^2$ ③, Similarly from Q to D:

$h - l\sqrt{3}/2 = \frac{1}{2} g \tau^2$ ④, From ③ & ④, $\frac{h}{h - l\sqrt{3}/2} = \frac{7}{4}$

$\Rightarrow h = 7\sqrt{3}l/6$ Ans.



Now $\tan\theta = \tan\alpha + \tan\alpha$
$= 2h/(R/2) = 4h/R$

$\Rightarrow \tan\theta = 4 \times \frac{7\sqrt{3}}{6} \times \frac{1}{\sqrt{7}}$

$\Rightarrow \tan\theta = \frac{2}{3}\sqrt{21} \Rightarrow \theta \approx 72°$.

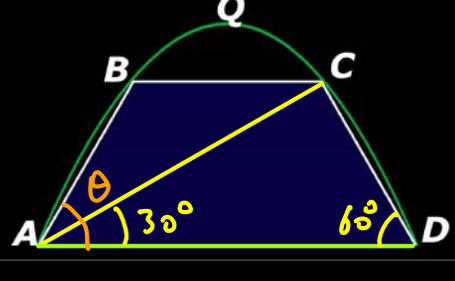
Now by energy conservation, $\frac{1}{2}mu^2 = mgh + \frac{1}{2}m(u\cos\theta)^2$

$\Rightarrow u = \frac{\sqrt{2gh}}{\sin\theta}$ where $h = \frac{7\sqrt{3}}{6} l$ and $\sin\theta = 2\sqrt{7/31}$

Thus, $u = \frac{1}{2}\sqrt{\frac{31gl}{\sqrt{3}}}$ Ans.

→ Think: What is the angle with horizontal at point A?

Solution: $\tan\theta = \tan 30° + \tan 60° = \frac{\sqrt{3} + 1}{\sqrt{3}}$
$\Rightarrow \theta = \tan^{-1}(4/\sqrt{3})$



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EXAMPLE (KVPY 2022 Q96):

Find the maximum possible angle of projection of a projectile such that its distance from launch point monotonically increases.

Solution: For a given launch angle all projectiles are similar since maximum height and range are both proportional to u^2.

Now recall that $y = x\tan\theta\left(1 - x/R\right)$, for simplicity we will find θ for a family of projectile for which $R = 1$. Now let r be distance from launch point. Then:

$r^2 = x^2 + y^2$, if r increases monotonically then, $\frac{d}{dx}(r^2) = 0$ should have no solution. Thus, $2x + 2y\frac{dy}{dx} = 0$ should have no solution, for $x > 0$.

Now $y = x\tan\theta\left(1 - \frac{x}{R}\right) = \tan\theta\left(x - x^2\right)$ for $R = 1$.

Thus $x + x\tan\theta\,(1-x)\times(\tan\theta)(1-2x) = 0$ should have no solution with $x > 0$.

i.e $\tan^2\theta\,(1-x)(1-2x) + 1 = 0$ should have no solution.

i.e $2x^2\tan^2\theta - 3x\tan^2\theta + (1 + \tan^2\theta) = 0$ should have no solution, thus its discriminant must be < 0.

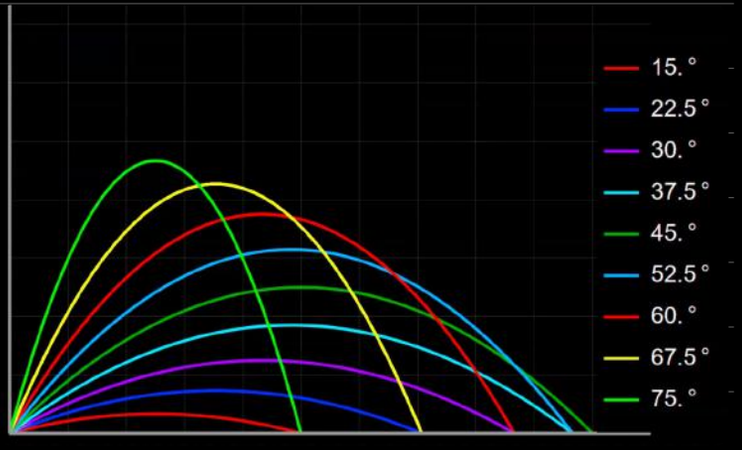
Thus $9\tan^4\theta - 8\tan^2\theta(1 + \tan^2\theta) = 0$ (in limiting case).

this gives $\tan\theta = 2\sqrt{2}$ $\Rightarrow$ Required solution $\theta < \tan^{-1}(2\sqrt{2})$

(We ignored the negative root and the trivial $\tan\theta = 0$).

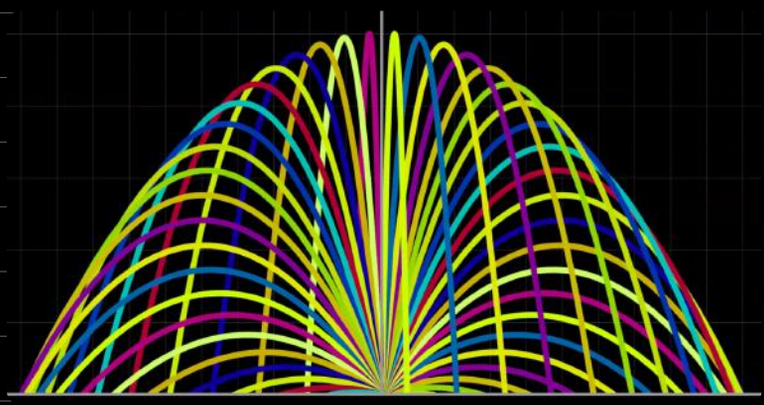
Thus $\theta < \tan^{-1}(2\sqrt{2})$ or alternatively $\theta < \cos^{-1}(1/3) \approx 70.5°$

→ For a visual idea, projectile paths for various launch angles are shown in adjacent figure. For 75° one can clearly see that distance from origin starts decreasing after a while while in all other cases it keeps increasing.



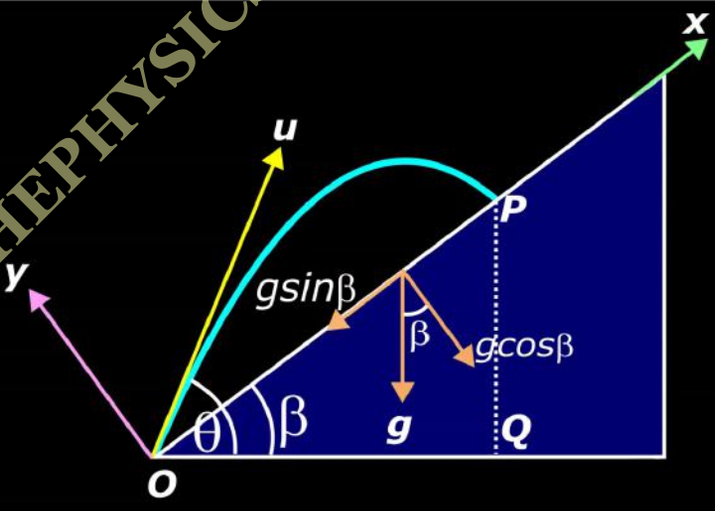
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→ The curves are a family of parabolae $y = x\tan\theta(1 - x/R)$, one can clearly see that ranges for complimentary angles are same.



→ A curve joining vertices of the family of parabolae is also a parabol (also known as safety parabola) we will later find its equation in a problem. If a bomb explodes uniformly then people outside outside this envelope are safe and hence the word safety parabola. (The Diwali "Anaar" has a similar behavior too).

Projection On An Inclined Plane:



Consider a projectile thrown upwards on an incline of inclination β. We choose x and y axes parallel and $\perp$ to incline. Let launching be with a speed u at an angle θ with horizontal.

Now resolving velocities and accelerations in to components:

$$u_x = u\cos(\theta-\beta),\quad u_y = u\sin(\theta-\beta)$$
$$a_x = -g\sin\beta \quad a_y = -g\cos\beta$$

Time Of Flight:

Now $s_y = u_y t + \frac{1}{2} a_y t^2$, @ $t = T$, $s_y = 0$

$$\Rightarrow \quad u\sin(\theta-\beta)T - \frac{1}{2} g\cos\beta\, T^2 = 0$$

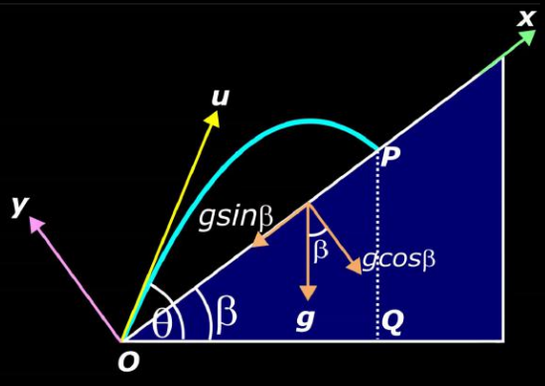
$$\Rightarrow \quad \boxed{T = \frac{2u\sin(\theta-\beta)}{g\cos\beta}}$$

→ This is similar to a stone thrown vertically upwards with substitutions

$$u \to u_y = u\sin(\theta-\beta) \quad a \to a_y = -g\cos\beta.$$

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RANGE:



$R = OP = OQ \sec\beta$. Now horizontal speed remains constant, $u_H = u\cos\theta$, thus, $OQ = u\cos\theta \times T$, thus:

$R = OQ\sec\beta = u\cos\theta \times T \times \sec\beta$, thus:

$$\boxed{R = \frac{2u^2 \sin(\theta-\beta)\cos\theta}{g\cos^2\beta}}$$

→ Range is same for θ and $(90°+\beta-\theta)$ (You can verify readily by substitution).

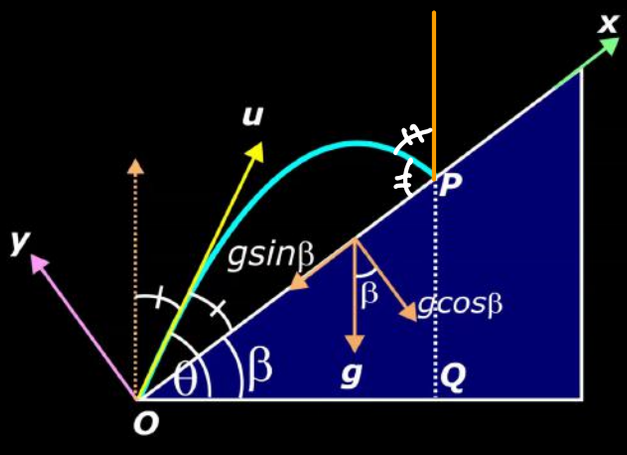
PROJECTION FOR MAXIMUM RANGE:

Using $2\sin A\cos B = \sin(A+B) + \sin(A-B)$, $R = \frac{u^2}{g\cos^2\beta}[\sin(2\theta-\beta) - \sin\beta]$

Thus for R_{max}, $\sin(2\theta-\beta) = 1 \Rightarrow \theta = \frac{\pi}{4} + \beta/2$ and correspondingly:

$$R_{max} = \frac{u^2}{g\cos^2\beta}[1-\sin\beta] = \frac{u^2(1-\sin\beta)}{g(1+\sin\beta)(1-\sin\beta)} = \frac{u^2}{g(1+\sin\beta)}$$

Thus $\boxed{R_{max} = \frac{u^2}{g(1+\sin\beta)}}$



For R_{max}, $\boxed{\theta = \pi/4 + \beta/2}$

→ This launching direction is bisector of vertical and the incline, i.e vertical and line $\vec{OP}$, this can be readily seen thus: Angle of u with vertical: $90° - (\frac{\pi}{4} + \beta/2) = 45° - \beta/2$
Angle of u with incline: $\frac{\pi}{4} + \beta/2 - \beta = 45° - \beta/2$ thus both angles are equal.

→ Landing speed is $\perp$ to launching speed when launched for maximum range this can be seen by dot product:

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$$\vec{u}\cdot\vec{v} = \vec{u}\cdot(\vec{u}+\vec{a}T) = u^2 + \vec{u}\cdot\vec{a}T$$

Now

$$\vec{u}\cdot\vec{a}T = u \times g \cos(90^\circ + 45^\circ + \beta/2)\cdot\frac{2u\sin(45^\circ-\beta/2)}{g\cos\beta}$$

$$= -\frac{2u^2\sin(45^\circ+\beta/2)\sin(45^\circ-\beta/2)}{\cos\beta} = -u^2$$

$(\because 2\sin A\sin B = \cos(A-B)-\cos(A+B))$

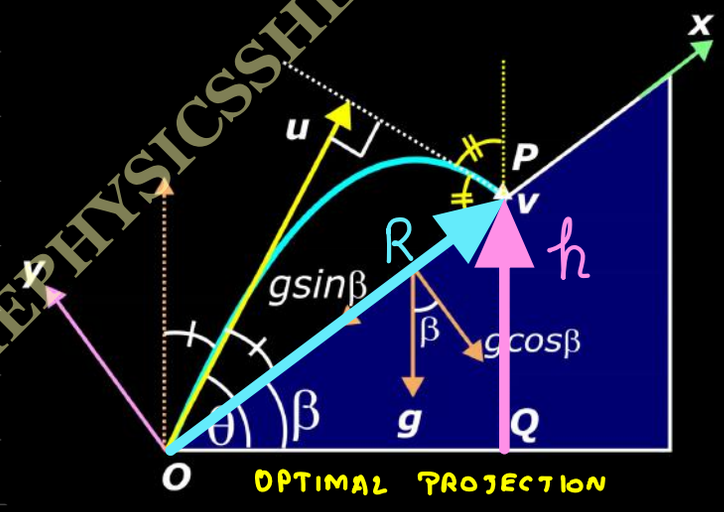
Thus $\vec{u}\cdot\vec{v} = 0$. QED.

Minimum Speed To Hit A Given Point:

If R is the maximum range corresponding to a certain u then that u is also the minimum speed for attaining that Range. This is because if that range were attainable with a smaller speed then range corresponding to this u will be longer.

Now at maximum range, $R = \frac{u^2}{g(1+\sin\beta)}$ as derived previously, rearranging this we get $u = \sqrt{g(R+R\sin\beta)}$, now from the figure we can see that $R\sin\beta$ is the vertical height of the landing point above the launching point. Thus minimum speed required to hit a point at a distance R and vertical height h:

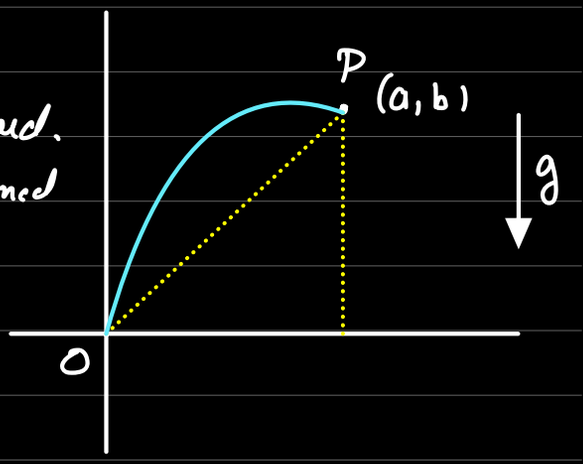
$$u_{min} = \sqrt{g(R+h)}$$



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→ This formula also hold true for points below the launching point. If the landing point is below the launching point, h is considered as negative. R is always considered positive.

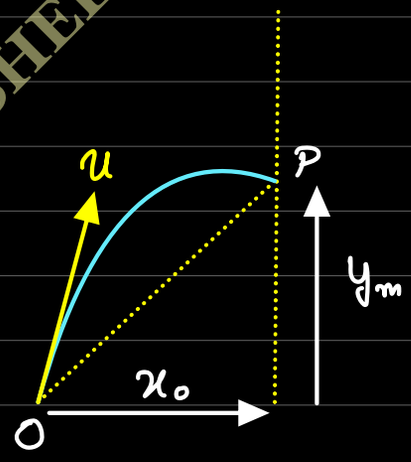
EXAMPLE: A stone is to be launched from origin to pass through point (a, b). Find the minimum required launching speed.



Solution: We can mentally imagine an inclined plane from origin to the given point. Now the problem is essentially same as that of finding minimum u to hit a point on given incline. Thus:

$$u_{min} = \sqrt{g(R+L)} = \sqrt{g(\sqrt{a^2+b^2})+b}$$ Ans.

EXAMPLE: A wall is located at a distance x_0 from the origin. Shots are fired towards the wall at various angles with the same speed u. Find the maximum achievable height y_m that can be achieved in this process.



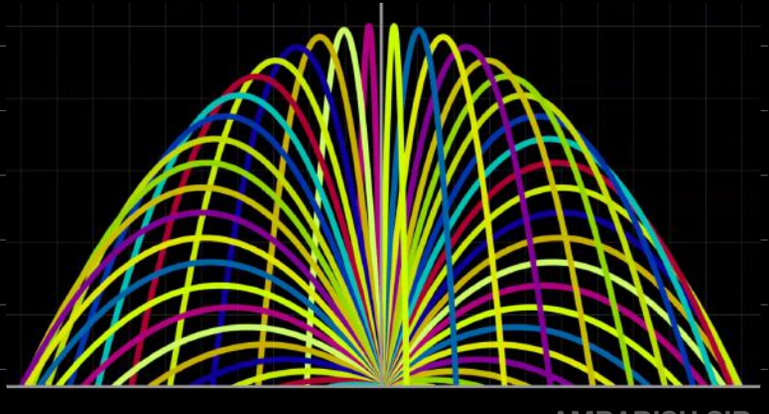
Solution: Imagine an incline joining the origin to the point of maximum achievable height. Now if y_m is maximum achievable height for a given u then that u is also the minimum required u to achieve that y_m. Thus: $u = \sqrt{g(\sqrt{x_0^2+y_m^2}+y_m)}$ $\left(\because \sqrt{g(R+L)}\right)$. Now rearranging this equation, we can solve for y_m. Thus:

$$\left(\frac{u^2}{g} - y_m\right)^2 = x_0^2 + y_m^2$$

Solving for y_m, we get $y_m = \frac{u^2}{2g} - \frac{1}{2}\frac{g x_0^2}{u^2}$

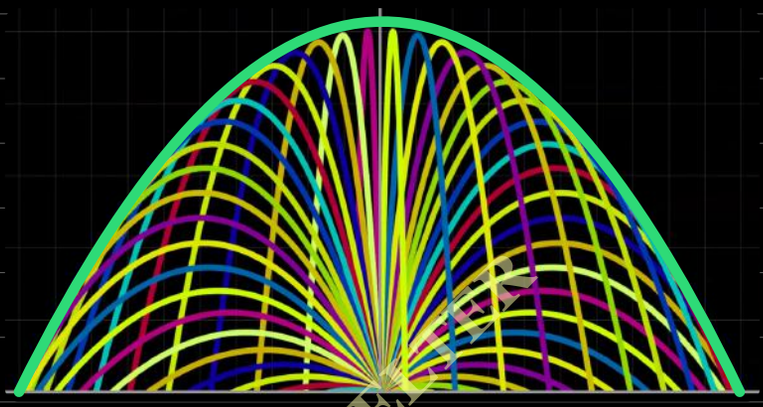
EXAMPLE: SAFETY ENVELOPE:

A fire cracker ("Anaar") sends out sparks and cinders isotropically with speed u. Find the equation of its safety envelope considering the location of the cracker as the origin, gravity in $-y$ direction.



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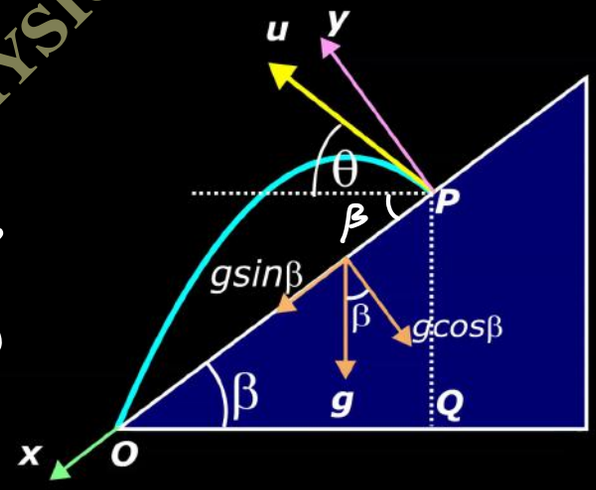
Solution: Let (x, y) be a point on safety envelope. Thus y must be the maximum achievable height for a given x. Therefore this problem is essentially same as the previous problem. We just need to substitute x for x_0 and y for y_m. Thus the required equation is:



$$y = \frac{u^2}{2g} - \frac{1}{2}\frac{gx^2}{u^2}$$

→ This happens to be a parabola, hence the word "safety parabola".

Projection Down The Incline:



Consider the adjacent figure, a particle is projected with speed u at an angle θ with horizontal. Resolving components,

$u_x = u\cos(\theta+\beta)$, $u_y = u\sin(\theta+\beta)$

$a_x = g\sin\beta$, $a_y = -g\cos\beta = -g\cos(-\beta)$

we see that u_x, u_y, a_x and a_y are obtained by simply substituting $\beta \to -\beta$ in the corresponding values for projection up the plane. Thus entire mathematics remains same for this case except for the substitution $\beta \to -\beta$. Therefore we can write all the formulae directly. Thus:

$$T = 2u\sin(\theta+\beta)/g\cos\beta$$

$$R = \frac{2u^2\sin(\theta+\beta)\cos\theta}{g\cos^2\beta}$$

$$R_{max} = \frac{u^2}{g(1-\sin\beta)}$$

For max Range, $\theta = 45° - \beta/2$

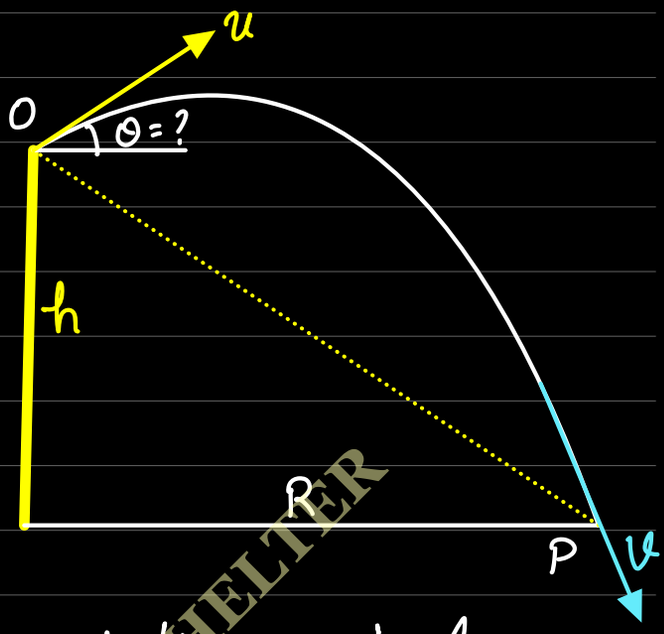
Range is same for θ and $(90° - \beta - \theta)$

→ Landing velocity is ⊥ to launching velocity for R_{max}

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→ For R_{max}, launching is at bisector of incline and vertical and landing is also along such bisector.

EXAMPLE: A projectile is to be launched from a tower of height h so as to land farthest. What should be the angle for such a projection. The launching speed is u.



Solution: We can mentally imagine an incline from launch point to the optimal landing point. For our problem, the range on this hypothetical incline should also be maximum. Because if the range on incline can be larger than OP, the trajectory will obviously cut the ground at a point farther than P. Thus we can say that landing velocity must be ⊥ to the launching velocity. Now since horizontal component of velocity always remains constant, therefore, $u\cos\theta = v\sin\theta$

$\Rightarrow$ $\tan\theta = u/v$, we can find v using energy conservation and our job is done.

$$\frac{1}{2}mu^2 + mgh = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{u^2+2gh}$$ thus:

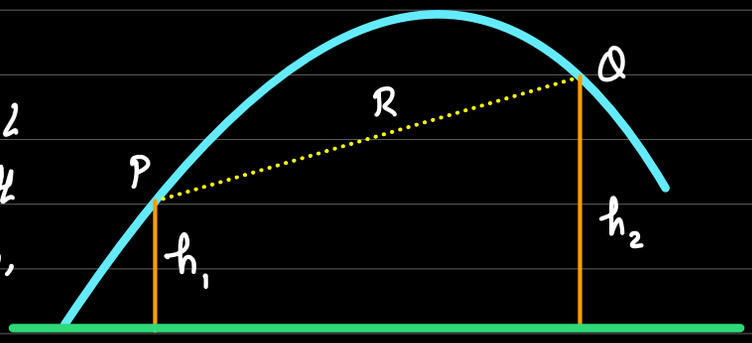
$$\boxed{\tan\theta = \frac{u}{\sqrt{u^2+2gh}}}$$

→ Similarly if projection point was at a lower level and landing point at a higher level, we could still use the same formula just that h will have to be replaced by $-h$, thus, for such cases, $\tan\theta = \frac{u}{\sqrt{u^2-2gh}}$, however in both cases we can always write $\boxed{\tan\theta = u/v}$

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Minimum Projection Speed To Pass Through Two Points:

Suppose we wish to throw a projectile from ground level so as to pass through two points P and Q as shown with minimum possible launch speed, then, the required minimum launching speed is given as:



$$\boxed{u_{min} = \sqrt{g(h_1 + h_2 + R)}}$$

This can be shown as follows:

Minimum speed is equivalent to minimum kinetic energy. Now according to our analysis for projection on inclined planes, we can write directly the minimum speed required at P for our objective as:

$$v_{p_{min}} = \sqrt{g(R + (h_2 - h_1))}$$

Now applying energy conservation from ground to P: $\frac{1}{2} m g (R + h_2 - h_1) + mg h_1 = \frac{1}{2} m u_{min}^2$

$$\Rightarrow \quad u_{min} = \sqrt{g(R + h_2 + h_1)} \quad \text{QED.}$$

→ After cancelling m, energy conservation equation can be directly written as $v^2 - u^2 = 2gh$ where h is the depth of point where speed is v (from the point where speed is u).

Path Of A Projectile As Seen From Another Projectile:

Suppose two bodies A and B are projected simultaneously from same point with speeds u_A and u_B and angles θ_A and θ_B respectively, then, the positions of two bodies are given as:

$$x_A = u_A \cos\theta_A t, \quad y_A = u_A \sin\theta_A t - \frac{1}{2} g t^2$$

$$x_B = u_B \cos\theta_B t, \quad y_B = u_B \sin\theta_B t - \frac{1}{2} g t^2$$

Now relative position is simply given as:

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$$\vec{r}_{BA} = (x_B - x_A)\hat{i} + (y_B - y_A)\hat{j} \text{ i.e}$$

$$\vec{r}_{BA} = (u_B \cos\theta_B - u_A \cos\theta_A)t\hat{i} + (u_B \sin\theta_B - u_A \sin\theta_A)t\,\hat{j}$$

Now $\vec{v}_{BA} = \frac{d}{dt}\vec{r}_{BA} = (u_B \cos\theta_B - u_A \sin\theta_A)\hat{i} + (u_B \sin\theta_B - u_A \sin\theta_A)\hat{j}$

Thus the relative velocity is constant, therefore B appears to move in a straight line along the direction given by $\phi = \tan^{-1}(v_{BA_y}/v_{BA_x})$
i.e $\phi = \tan^{-1}[(u_B \sin\theta_B - u_A \sin\theta_A)/(u_B \cos\theta_B - u_A \cos\theta_A)]$

→ Further if $u_B \sin\theta_B = u_A \sin\theta_A$ i.e vertical components are equal then $\phi = 0$ i.e B appears to move along a horizontal line.

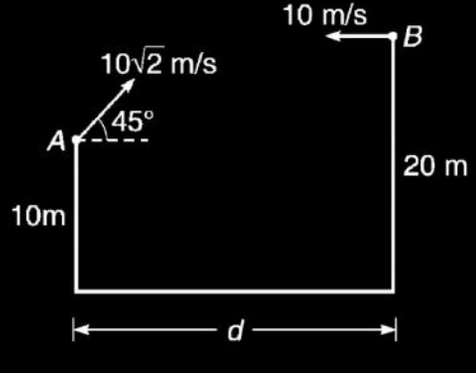
→ If $u_A \cos\theta_A = u_B \cos\theta_B$ i.e horizontal components are equal then $\phi = 90°$, i.e B appears to move vertically wrt A.

→ If the launch points were not same, still relative velocity expression will remain the same even though the relative position expression will be offset by initial relative position.

→ For collision of projectiles, the relative motion of the projectiles should be in exact opposite direction as initial relative position (i.e from frame of one projectile, the other one must appear to approach head on for a collision).

EXAMPLE:

Two particles A and B are projected simultaneously from two towers of heights 10 m and 20 m respectively. Particle A is projected with an initial speed of $10\sqrt{2}$ m/s at an angle of 45° with horizontal, while particle B is projected horizontally with speed 10 m/s. If they collide in air, what is the distance d between the towers?



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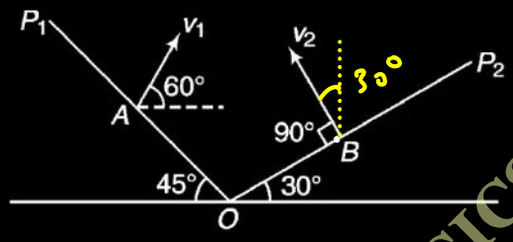
Solution: $\vec{V}_{BA} = \vec{V}_B - \vec{V}_A = -10\hat{i} - (10\hat{i} + 10\hat{j}) = -20\hat{i} - 10\hat{j}$

$\Rightarrow$ direction of travel : $\phi = 180° + \tan^{-1}(10/20) = 180° + \tan^{-1}(1/2)$

Thus relative position $\vec{r}_{BA}$ must be along $\tan^{-1}(1/2)$, Now relative position: $\vec{r}_{BA} = d\hat{i} + 10\hat{j}$, $\therefore \frac{10}{d} = \frac{1}{2} \Rightarrow d = 20\,m$ Ans.

EXAMPLE :

A particle is projected from an inclined plane OP_1 from A with velocity $v_1 = 8\ ms^{-1}$ at an angle 60° with horizontal. An another particle is projected at the same instant from B with velocity $v_2 = 16\ ms^{-1}$ and perpendicular to the plane OP_2 as shown in figure. After time $10\sqrt{3}$ s their separation was minimum and found to be 70 m. Then find distance AB.



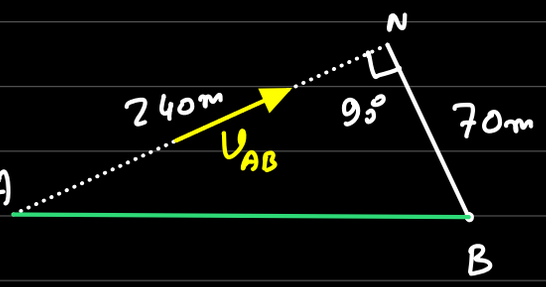
Solution: From B's reference frame, A appears to move along a straight line. Now, U_1 is inclined 30° clockwise from vertical while V_2 is inclined 30 anticlockwise from vertical, thus angle between them is 60°. Thus,

$|V_{AB}| = \sqrt{8^2 + 16^2 - 2\times8\times16\cos60°} = 8\times\sqrt{1+4-2} = 8\sqrt{3}\ ms^{-1}$. Thus, displacement of A as seen from B in $10\sqrt{3}$ sec $S = 8\sqrt{3}\times10\sqrt{3} = 240\,m$.

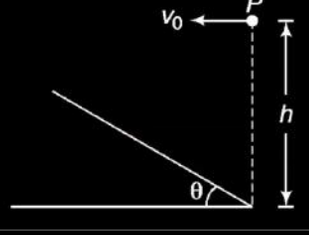
Now let N be the nearest distance point. Based on information and calculations, we draw the adjacent figure.

Now by pythagoras theorem, $AB = \sqrt{70^2 + 240^2}$

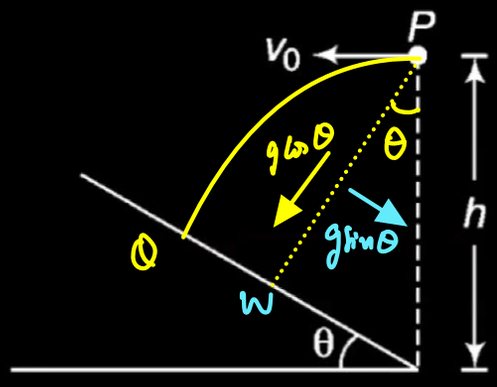
$\Rightarrow$ $AB = 250\,m$ Ans.



EXAMPLE : Determine the horizontal velocity v_0 with which a stone must be projected horizontally from a point P, so that it may hit the inclined plane perpendicularly. The inclination of the plane with the horizontal is θ and point P is at a height h above the foot of the incline, as shown in the figure.



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Solution: Let T be the time of flight. During this time, component of velocity ⊥ to incline must vanish and the particle should travel a distance $h\cos\theta$ along direction PW. Converting this information to equations:

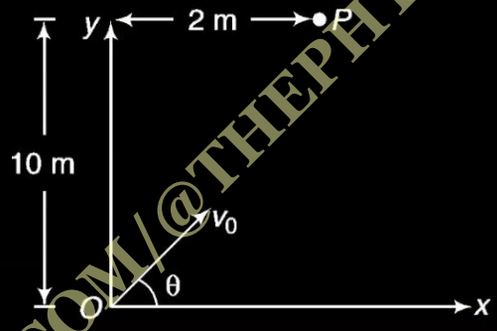
$$v_0\cos\theta - g\sin\theta\, T = 0 \quad (1)$$

$$v_0\sin\theta\, T + \frac{1}{2}g\cos\theta\, T^2 = h\cos\theta \quad (2)$$

Solving (1) & (2) $\quad v_0 = \sqrt{2gh/(2+\cot^2\theta)}$ Ans.

EXAMPLE:

A particle is dropped from point P at time $t = 0$. At the same time another particle is thrown from point O as shown in the figure and it collides with the particle P. Acceleration due to gravity is along the negative y-axis. If the two particles collide 2 s after they start, find the initial velocity v_0 of the particle which was projected from O. Point O is not necessarily on ground.

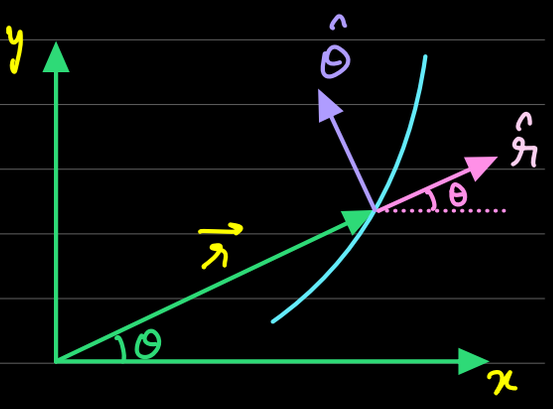


Solution: This is standard problem in collision of projectiles. Now at the initial moment, $\vec{r}_{po} = 2\hat{i} + 10\hat{j}$ i.e along $\tan^{-1}5$ direction. For hitting, $\vec{v}_{op}$ must be along this direction. Thus, $\tan\theta = 5$. Now for hitting after 2 seconds, The horizontal coordinates of two projectiles must be equal at this instant, thus, $v_0\cos\theta \times 2 = 2$

$\Rightarrow v_0 = 1/\cos\theta$, $\cos\theta = 1/\sqrt{26}$, thus $v_0 = \sqrt{26}$ m/s Ans.

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Unit Vectors And Their Derivatives In Polar Coordinates:



Recall that we had introduced polar coordinate system during our study of vectors. Consider a particle located at $r\angle\theta$ wrt origin. A unit vector along the $\vec{r}$ is called $\hat{r}$, as can be readily seen from the figure,

$\hat{r} = \cos\theta\, \hat{i} + \sin\theta\, \hat{j}$ ①, this is also known as radial unit vector. A unit vector $\perp$ to $\hat{r}$ in the anticlockwise direction is called tangential unit vector of the polar coordinate system $\hat{\theta}$ (not to be confused with tangential vector along path). From the figure above, it can be readily seen that $\hat{\theta} = -\sin\theta\, \hat{i} + \cos\theta\, \hat{j}$ ②

Now as the motion of the particle progresses, r and θ keep changing with time and accordingly $\hat{r}$ and $\hat{\theta}$ keep changing with time.

Now differentiating ① wrt time, we get:

$$\dot{\hat{r}} = -\sin\theta\,\dot{\theta}\,\hat{i} + \cos\theta\,\dot{\theta}\,\hat{j} = \omega\,\hat{\theta} \quad \text{where } \omega = \dot{\theta}$$

thus:

$$\boxed{\dot{\hat{r}} = \omega\,\hat{\theta}} \quad ③$$

Now Differentiating ② wrt time we get:

$$\dot{\hat{\theta}} = -\cos\theta\,\dot{\theta}\,\hat{i} - \sin\theta\,\dot{\theta}\,\hat{j} = -\omega\,\hat{r} \quad \text{thus:}$$

$$\boxed{\dot{\hat{\theta}} = -\omega\,\hat{r}} \quad \text{where } \omega = \dot{\theta}$$

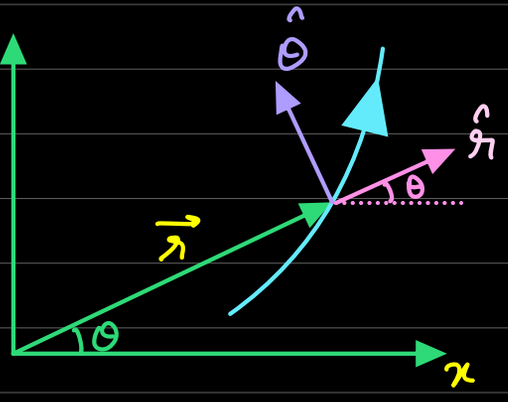
$\vec{\omega}$ of A Moving Particle About Origin:

If a particle located at $r\angle\theta$ is moving in space such that θ is changing at a rate $\dot{\theta}$ then we define $\vec{\omega}$ as a vector whose magnitude is $|\dot{\theta}|$ and direction is given by right hand thumb rule, i.e curl the fingers of right hand in sense of rotation of the particle and thumb gives the direction of $\vec{\omega}$.

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For instance for a particle moving along the turquoise path, vector joining origin to the particle is rotating anti-clockwise. Now curling the fingers of right hand in the anti-clockwise direction while keeping the thumb ⊥ to the plane of motion, the thumb must point outwards (along $\hat{k}$) thus $\vec{\omega}$ of this particle about the origin is outwards from the plane of the page and its magnitude is $\dot{\theta}$.

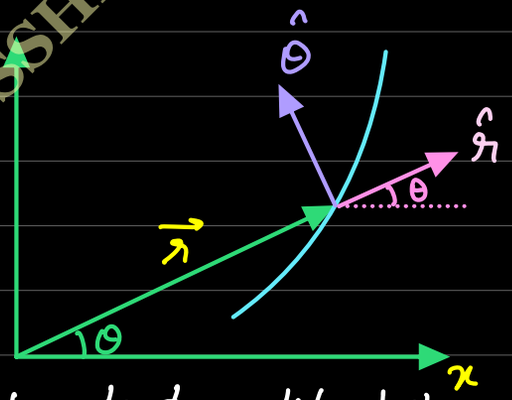


Velocity Vector In Polar Coordinates:

Position of any particle in polar coordinates can be written as $\vec{r} = r\hat{r}$, thus velocity of the particle:

$$\vec{v} = \dot{\vec{r}} = r\dot{\hat{r}} + \dot{r}\hat{r}$$

$$\boxed{\vec{v} = r\omega\hat{\theta} + \dot{r}\hat{r}}$$



Thus $\vec{v} = v_\tau\hat{\theta} + v_r\hat{r}$ where $v_\tau = \omega r$ represents tangential velocity and $v_r = \dot{r}$ represents radial component of velocity. If suppose particle were moving with r = constant (say arc of a circle) then, $\dot{r} = 0$ and $\vec{v} = r\omega\hat{\theta}$ which could also be vectorially written as

$$\boxed{\vec{v} = \vec{\omega} \times \vec{r}}$$

The student can readily verify that $\vec{\omega} \perp \vec{r}$ thus $|\vec{\omega} \times \vec{r}| = \omega r$ and direction of $\vec{\omega} \times \vec{r}$ is in the required tangential direction. The vector formula works irrespective of the fact whether particle is rotating clockwise or anti-clockwise because $\vec{\omega}$ appropriately changes the direction according to sense of rotation of the particle.

$\vec{\alpha}$ of A Moving Particle About Origin: In a previous section we defined $\vec{\omega}$ of a moving particle (also known as angular velocity). $\vec{\alpha}$ (also known as angular acceleration vector) is simply rate of

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change of $\vec{\omega}$. If a particle's tangential speed is increasing, then $\vec{\alpha} \parallel \vec{\omega}$ while if the rate of rotation of a particle is decreasing then $\vec{\alpha} \nparallel \vec{\omega}$. For example when you switch on a fan, its $\vec{\omega}$ as well as $\vec{\alpha}$ point downwards while when you switch it off, $\vec{\omega}$ points downwards while $\vec{\alpha}$ points upwards. $\because$ magnitude of ω is $\dot{\theta}$, magnitude of $\vec{\alpha}$ is $\ddot{\theta}$. For motion in the plane of the page, if $\ddot{\theta}$ is positive, $\vec{\alpha}$ is coming out of the page and if $\ddot{\theta}$ is negative, $\vec{\alpha}$ is in to the page.

Acceleration Of A Particle In Polar Coordinates:

In a previous section we showed that:

$\vec{v} = \omega r \hat{\theta} + \dot{r}\hat{r}$, Now $\vec{a}$ is simply $\dot{\vec{v}}$ thus,

$$\vec{a} = \dot{\vec{v}} = \omega r \dot{\hat{\theta}} + \omega \dot{r} \hat{\theta} + \dot{\omega} r \hat{\theta} + \ddot{r}\hat{r} + \dot{r}\dot{\hat{r}}$$

($\because$ it can be shown by repeated product rule that $\frac{d}{dt}(xyz) = \dot{x}yz + x\dot{y}z + xy\dot{z}$).

Now $\dot{\hat{r}} = \omega\hat{\theta}$ and $\dot{\hat{\theta}} = -\omega\hat{r}$ thus,

$\vec{a} = -\omega^2 r\hat{r} + \omega\dot{r}\hat{\theta} + \dot{\omega} r\hat{\theta} + \dot{r}\hat{r} + \omega\dot{r}\hat{\theta}$ thus:

$$\boxed{\vec{a} = (-\omega^2 r + \ddot{r})\hat{r} + (\alpha r + 2\omega\dot{r})\hat{\theta}}$$

Vectorially this can be written as:

$$\boxed{\vec{a} = [\vec{\omega} \times (\vec{\omega} \times \vec{r})] + \ddot{r}\hat{r} + (\vec{\alpha} \times \vec{r}) + [2\vec{\omega} \times \vec{v}_{rot}]}$$

Here $\vec{v}_{rot}$ represents velocity vector of the particle in a frame in which particle appears to move only in radial direction (i.e a frame rotating with $\dot{\theta}$).

→ The term $2\vec{\omega} \times \vec{v}_{rot}$ is called coriolis acceleration.

→ If particle is moving at a constant radius (i.e circular motion),

$\vec{a} = -\omega^2 r\hat{r} + r\alpha\hat{\theta}$, the term $-\omega^2 r\hat{r}$ is called centripetal acceleration and $r\alpha$ is called tangential acceleration.

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Vectorially in circular motion:

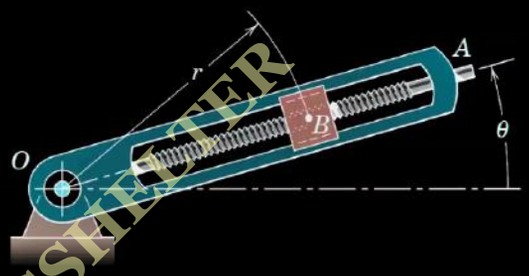
$$\vec{a} = \vec{\omega} \times (\vec{\omega} \times \vec{r}) + \vec{\alpha} \times \vec{r}$$

$\vec{a}_c = \vec{\omega} \times (\vec{\omega} \times \vec{r})$ ← Centripetal acceleration, $\vec{a}_t = \vec{\alpha} \times \vec{r}$ ← Tangential acc^n.

→ To avoid confusion θ must be considered positive in the anti-clockwise direction only or else vector formulas must be used to avoid any oversight.

EXAMPLE:

Rotation of the radially slotted arm is governed by $\theta = 0.2t + 0.02t^3$, where θ is in radians and t is in seconds. Simultaneously, the power screw in the arm engages the slider B and controls its distance from O according to $r = 0.2 + 0.04t^2$, where r is in meters and t is in seconds. Calculate the magnitudes of the velocity and acceleration of the slider for the instant when $t = 3$ s.



Solution: $\vec{v} = \omega r \hat{\theta} + \dot{r}\hat{r} \Rightarrow v = \sqrt{\omega^2 r^2 + \dot{r}^2}$

Now $\omega = \dot{\theta} = 0.2 + 0.06t^2$

$\dot{r} = 0.08t$, Now at $t = 3$ sec

$r = 0.2 + 0.04 \times 3^2 = 0.56$ m

$\dot{r} = 0.08 \times 3 = 0.24$ m s^{-1}

$\omega = 0.2 + 0.06 \times 3^2 = 0.74$ rad s^{-1} thus,

Thus radial velocity, $v_r = \dot{r} = 0.24$ m s^{-1}

Tangential velocity $v_\tau = \omega r = 0.56 \times 0.74 = 0.414$ m s^{-1}

Thus speed $= \sqrt{v_r^2 + v_\tau^2} = \sqrt{0.24^2 + 0.414^2} = 0.479$ m s^{-1}

Now,

$\ddot{r} = 0.08$

$\alpha = \ddot{\theta} = \dot{\omega} = 0.12t$ @ $t = 3$, $\ddot{\theta} = 0.36$ rad s^{-2}

Now acceleration in radial direction: $a_r = \ddot{r} - \omega^2 r = 0.08 - 0.56 \times 0.74^2$

$\Rightarrow a_r = -0.227$ m s^{-2}

$a_\theta = r\alpha + 2\omega\dot{r} = 0.56 \times 0.36 + 2 \times (0.24)(0.74) = 0.557$ m s^{-2}

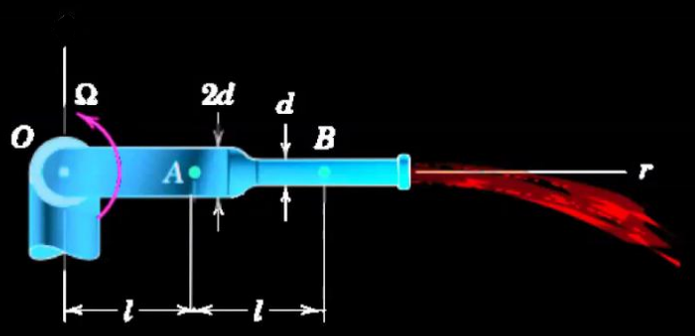
Thus

$$a = \sqrt{a_r^2 + a_\theta^2} = \sqrt{0.227^2 + 0.557^2} = 0.601 \text{ m s}^{-2} \text{ Ans.}$$

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EXAMPLE:

The nozzle shown rotates with constant angular speed Ω about a fixed horizontal axis through point O. Because of the change in diameter by a factor of 2, the water speed relative to the nozzle at A is v, while that at B is $4v$. The water speeds at both A and B are constant. Determine the velocity and acceleration of a water particle as it passes (a) point A and (b) point B.



Solution: For A, $\dot{r} = v \quad \dot{\theta} = \Omega \Rightarrow v_A = \sqrt{(\Omega l)^2 + v^2}$ Ans.

For B, $\dot{r} = 4v \quad \dot{\theta} = \Omega \Rightarrow v_B = \sqrt{(\Omega \times 2l)^2 + (4v)^2}$ Ans

Now $\vec{a} = (\ddot{r} - \omega^2 r)\hat{r} + (r\alpha + 2\omega\dot{r})\hat{\theta}$, Here $\ddot{r} = 0, \alpha = 0$

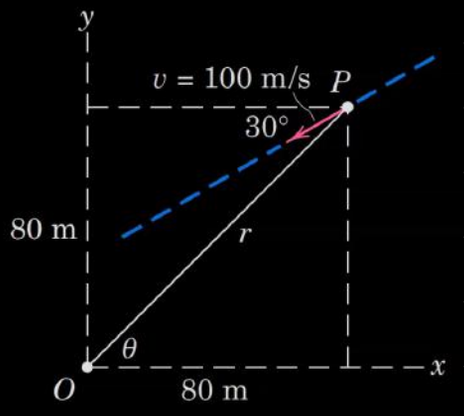
$\Rightarrow \vec{a} = -\omega^2 r \hat{r} + 2\omega\dot{r}\hat{\theta}$

thus $\vec{a}_A = -\Omega^2 l \hat{r} + 2\Omega v \hat{\theta} \qquad \vec{a}_B = -2\Omega^2 l \hat{r} + 2\Omega(4v)\hat{\theta}$

$\Rightarrow \vec{a}_B = -2\Omega^2 l \hat{r} + 8\Omega v \hat{\theta}$ Ans.

EXAMPLE:

As it passes the position shown, the particle P has a constant speed $v = 100$ m/s along the straight line shown. Determine the corresponding values of $\dot{r}$, $\dot{\theta}$, $\ddot{r}$, and $\ddot{\theta}$.



Solution: Here $\theta = 45°$, $\dot{r} = -100\cos 15°$ Ans.

$v_\tau = 100\sin 15° = 80\sqrt{2}\,\dot{\theta} \Rightarrow \dot{\theta} = 100\sin 15°/80\sqrt{2} = \frac{5}{4\sqrt{2}}\sin 15°$

Now Since speed is constant, $a_r = 0$, $a_\tau = 0$

$\Rightarrow \ddot{r} - \omega^2 r = 0 \Rightarrow \ddot{r} = \omega^2 r = 80\sqrt{2} \times \left(\frac{5}{4\sqrt{2}}\sin 15°\right)^2 = \frac{125\sin^2 15°}{\sqrt{2}}$ Ans.

$a_\tau = r\ddot{\theta} + 2\omega\dot{r} \Rightarrow \ddot{\theta} = -2\omega\dot{r}/r = -2 \times \frac{5}{4\sqrt{2}}\sin 15° \times (-100\cos 15°) \times \frac{1}{80\sqrt{2}}$

$= \frac{25}{8}\sin 30° = 25/16$ rad s^{-2} Ans.

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EXAMPLE: A particle moves outwards along a spiral $r = \theta/\pi$ where r is in metres and θ is in radians (angular coordinate). θ varies with time as $\theta = \alpha t^2/2$. (a) When is the radial acceleration of particle zero (b) At what θ radial and tangential accelerations have equal magnitudes.

Solution: $a_r = \ddot{r} - \omega^2 r$, $r = \frac{1}{\pi}\cdot\left[\frac{\alpha t^2}{2}\right] \Rightarrow \dot{r} = \frac{\alpha t}{\pi}$, $\ddot{r} = \frac{\alpha}{\pi}$

$\omega = \dot{\theta} = \alpha t$ thus $a_r = \frac{\alpha}{\pi} - (\alpha t)^2 \times \frac{\alpha t^2}{2\pi} = 0$ (for part a)

$\Rightarrow \alpha^3 t^4 = 2\alpha \Rightarrow t = (2/\alpha^2)^{1/4}$ sec.

Now $a_t = r\dot{\omega} + 2\omega\dot{r} = \frac{\alpha t^2}{2\pi} \times \alpha + 2\alpha t \times \frac{\alpha t}{\pi} = \frac{5}{2}\frac{\alpha^2 t^2}{\pi}$

Now $a_t = |a_r|$ for part b thus $\frac{5}{2}\frac{\alpha^2 t^2}{\pi} = \left|\frac{\alpha}{\pi} - \frac{\alpha^3 t^4}{2\pi}\right| \Rightarrow$ Either:

$\Rightarrow \alpha^3 t^4 + 5\alpha^2 t^2 - 2\alpha = 0 \Rightarrow t = \sqrt{\frac{\sqrt{33} - 5}{2\alpha}}$ sec

or $\alpha^3 t^4 - 5\alpha^2 t^2 - 2\alpha = 0 \Rightarrow t = \sqrt{\frac{\sqrt{33} + 5}{2\alpha}}$ sec.

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